

Raw Input Table

Name	Assumed office	Departed office	Political party	Notes/Events
Benjamin Contee	March 4, 1789	March 3, 1791	Anti-Administration	Resigned
William Pinkney	March 4, 1791	November 1791	Pro-Administration	
Benjamin Cardin	January 3, 1987	January 3, 2007	Democratic	
John Sarbanes	January 3, 2007	Incumbent	Democratic Party	
...	...	...	...	...

- ✗

Date fields are strings
- ✗

Names not decomposed
- ✗

Term duration not explicit
- ✗

Inconsistent party labels

Q: What is the **average duration** of representatives in Democratic Party?
Q: Which representatives of Democratic Party assumed office **after 1980**?

✗----- SQL fails on string dates -----✗

Before transformation

QuIeTT Pipeline

Stage 1 - Issue Probing

Synthetic Query Generation

- How long was Benjamin Cardin in office?
- Who served for more than two terms?

Table Issues Exposed

- Date fields are strings
- Inconsistent party labels
- Implicit end dates (Incumbent)
- Term duration not explicit
- Names not decomposed

Probe questions discarded, only issues kept

Stage 2.1 - Plan Generation

JSON Plan Generation

- Row identity & raw data preserved
- Canonicalize textual placeholders
- Parse free-text dates

Transformation Ops

- Derived column
- Normalization
- Restructuring

Stage 2.2 - Plan Execution

Code Generation

Plan → Code

Code Execution

Materialise canonical table

Transformed Table

First Name	Last Name	Start Date	End Date	Start Year	End Year	Term Duration	Political Affiliation	Notes/Events
Benjamin	Contee	1789-03-04	1791-03-03	1789	1791	2.0	Anti-Administration	Not available
William	Pinkney	1791-03-04	1791-11-01	1791	1791	0.0	Pro-Administration	Resigned
Benjamin	Cardin	1987-01-03	2007-01-03	1987	2007	20.0	Democratic	Not available
John	Sarbanes	2007-01-03		2007			Democratic	Not available
...	...	...	...	...	...	...	...	...

- ✓

Dates: YYYY-DD-MM
- ✓

Canonical version
- ✓

Duration
- ✓

NaN

Transform once per-table

Stage 3 - Query Time

Q: What is the **average duration** of representatives in Democratic Party?
A: 20.0

Q: Which representatives of Democratic Party assumed office **after 1980**?
A: Benjamin Cardin and John Sarbanes

Question 3
COT → SQL → answer

Question n
COT → SQL → answer

Query n times